

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI, PILANI

FIRST SEMESTER 2023– 2024

CE F231 (FLUID MECHANICS) EVALUATIVE-3 (OPEN BOOK)

Duration: 45 mins

Max Marks: 20

Note: Please write the answers in SI units and up to three decimal points with units. Use $g=9.81 \text{ m/s}^2$.

- Q1.** A solid cylinder of diameter 4.5 m has a height of 2.5 m. Find the meta-centric height of the cylinder when it is floating in water with its axis vertical. The specific gravity of the cylinder is 0.45. [6]
- Q2.** A stone weighs 450 N in air and 250 N in water. Compute the volume of the stone. What would it weigh in mercury with specific gravity 13.6? [4]
- Q3.** Find the density of metallic body which floats at the interface of mercury of specific gravity 13.6 and water such that 40% of its volume is submerged in mercury and 60% in water. [4]
- Q4.** A plane flow has velocity components $u = \frac{x}{T_1}$; $v = -\frac{y}{T_2}$; and $w = 0$ along x, y, and z direction respectively, where $T_1 \neq 0$ and $T_2 \neq 0$ are constant having the dimension of time. What is the relationship between T_1 and T_2 if the flow is incompressible? [2]
- Q5.** For a two-dimensional flow field, the stream function $\psi = \frac{3}{2}(y^2 - x^2)$. What is the magnitude of discharge occurring between the streamline passing through points (0,3) and (3,4)? Assume unit width and mention discharge in units. [2]
- Q6.** The velocity field for flow is given by,
 $\vec{V} = (5x + 6y + 7z) \hat{i} + (6x + 5y + 9z) \hat{j} + (3x + 2y + \lambda z) \hat{k}$ and the density varies as $\rho = \rho_0 e^{-2t}$. In order that mass is conserved, what should be the value of λ ? [2]